

ChipChat Assignment

Conversational hardware design with ChipChat

Course: LLM4ChipDesign | Submission: GitHub repo only | Updated version

Overview

In this assignment, you will work through the provided **ChipChat.ipynb** notebook and use the ChipChat workflow to iteratively generate and refine hardware designs through natural-language conversation. This updated version of the assignment now requires you to complete **all eight examples** in the notebook, as well as the additional **CPU example** shown in the figure below.

Learning goals

- Understand how interactive human feedback affects the success of LLM-based hardware design.
- Gain hands-on experience with a conversational chip-design workflow across multiple examples.
- Compare the behavior of the method on smaller examples versus a more complex CPU-style design.
- Reflect on the strengths and limitations of ChipChat for design generation.

Free chat

- Providing human feedback (the “conversation”) increased success
- Can we leverage this for more complex designs?
- Let’s try make an (8-bit) processor

```
1 Let us make a brand new microprocessor design together. We're severely constrained on space and I/O. We have to fit in 1000 standard cells of an ASIC, so I think we will need to restrict ourselves to an accumulator based 8-bit architecture with no multi-byte instructions. Given this, how do you think we should begin?
```

Fig. 10. 8-bit accumulator-based processor: Starting co-design prompt

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Figure: CPU / 8-bit processor example to be completed in addition to the eight notebook examples.

Environment and setup

- Use the provided **ChipChat.ipynb** notebook in Google Colab or a local Jupyter environment.
- Install any required dependencies and configure the needed API key if the notebook requires one.
- The notebook should run from top to bottom, except for the designated API-key insertion step.
- Do **not** include API keys or secrets in your repository.

Required tasks

Please complete the following parts and document your results clearly.

Part I - Complete all eight examples in the notebook

- Run and complete **all eight examples** included in **ChipChat.ipynb**.
- For each example, preserve the prompts / conversations / outputs that illustrate the ChipChat workflow.
- For each example, keep the final generated design artifacts and any simulation / compilation / textual results produced.
- If the notebook asks for analysis or follow-up discussion, answer those items clearly.

Part II - Complete the CPU example

- Use the conversational ChipChat methodology to work through the **CPU / 8-bit processor** example shown in the figure.
- Document the initial prompt and the subsequent interaction used to develop the design.
- Show the final generated design result and summarize the important architectural choices.
- Include any available outputs, code, or evidence of the completed workflow.

Part III - Reflection and comparison

- Compare the CPU example with the eight smaller examples.
- Comment on whether human feedback seems especially helpful for more complex designs.
- Identify at least two strengths and two limitations of the ChipChat workflow.
- Mention any examples where the generated result needed significant correction.

What to submit

Submission method: **GitHub repository only** (no email submission). Please create or use your homework repository and organize this assignment in a folder such as **HW_ChipChat/**.

- **Executed notebook:** a completed notebook file, for example **ChipChat_hw.ipynb**, containing all eight examples and the CPU example.
- **Generated artifacts:** any generated Verilog/code/logs/output files from the notebook runs.
- **Report PDF:** **ChipChat_Report.pdf** summarizing your results and reflections.
- **Repo access:** invite **weihuax6@gmail.com** as a collaborator on your repository.

Report requirements (ChipChat_Report.pdf)

- List all eight completed notebook examples.
- Provide a short summary for each example.
- Provide a more detailed description of the CPU example, including the starting prompt and final result.
- Discuss how the conversation / human feedback influenced the design process.
- Briefly note any failures, debugging steps, or improvements across iterations.

Grading rubric

Criterion	Points
Completion and correctness of all 8 notebook examples	40
Completion and discussion of the CPU example	25
Quality of analysis and reflection	15
Organization and completeness of submitted files	10
Clarity, reproducibility, and report quality	10

Important notes

- Make sure notebook outputs are visible and clearly labeled.
- Use meaningful file and folder names in your repository.
- Do not submit any API keys, credentials, or other private information.
- Unless otherwise specified, the submitted work should be your own.